

DNA and Genetics Basics

DNA stores hereditary information used by cells to build and regulate living systems.

Key facts:

DNA is a polymer made of nucleotides containing a sugar, phosphate group, and nitrogenous base.

The four DNA bases are adenine, thymine, cytosine, and guanine.

Base-pairing rules are adenine with thymine and cytosine with guanine.

A gene is a DNA sequence that contributes to a functional product, often a protein or functional RNA.

DNA replication copies genetic information before cell division.

Transcription copies information from DNA into RNA, and translation uses messenger RNA to build proteins.

Mutations are changes in DNA sequence; their effects can be harmful, neutral, or beneficial depending on context.

Answerable questions:

Q: What does DNA store? A: DNA stores genetic information in the sequence of its nucleotide bases.

Q: What are the base-pairing rules? A: A pairs with T, and C pairs with G in double-stranded DNA.